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## Experiment - 3 Study of Network Cables and Crimping with RJ45 Connector

**Aim:** To study of different types of network cables and crimping with RJ45 connector

**Content:**

| Cable Type | Category | Maximum data transmission | Application |
|---|---|---|---|
| UTP | Category 3 | 10 bps | 10 BaseT |
| | Category 5 | upto 100 Mbps | Fast, Gigabit |
| | Category 5e | 1 Gbps | Fast, Gigabit |
| STP | Category 6, 6a | 10Gbps | Gigabit, 10G widely used in data centers |
| SSTP | Category 7 | 10Gbps | Gigabit, 10G (100m) |
| Coaxial Cable | RG-6, RG-59 RG-11 | 10-100Mbps | Speed of signal is 500m, Television network. |
| Fibre Optic Cable | Single Mode Multi Mode | 100 Gbps | Maximum distance of fibre optic cable. |

RJ45 is the standard connector for Ethernet cables. Two wiring standards exist :- TIA/EIA 568 A : Green - white, Green, Orange - white, Blue, Blue - White, Orange, Brown - white, Brown - TIA/EIA 568 B : Orange - white, Orange, Green - white, Blue, Blue - White, Green, Brown - white, Brown.

Straight - through cable uses the same standard (568 B - 568 B) on both ends - connects PC to switch. Cross - over cable uses 568 A on one end and 568 B on the other connects PC to PC directly.

Result :

Both straight - through and Cross - over Ethernet cables were successfully crimped and verified using a cable tester. All 8 pins showed proper continuity.